Groq Tensor Streaming Multiprocessor

A complete overview on Groq and Comparative Study with Google TPU

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***Abstract*—Deep learning and its applications have grown rapidly, which has increased demand for specialised hardware designs that can speed up computationally demanding activities like training and inference. To meet the particular problems presented by these workloads, Silicon Valley startup Groq has created a revolutionary processing architecture known as the Tensor Streaming Multiprocessor (TSP). An extensive study of the TSP architecture, its main advances, and its possible effects on artificial intelligence (AI) are presented in this work. Furthermore, a comparison is made between Google's Tensor Processing Unit (TPU), another well-known specialised AI accelerator, and Groq's TSP.**

Index Terms*—*Tensor Streaming Multiprocessor, TSM, Deep Learning, AI Accelerators, Specialized Hardware,Google Tensor , Groq Chip, Tensor Processing Unit, TPU

I. Introduction

Artificial intelligence (AI), and deep learning in particular, has advanced dramatically in the last few years, transforming a number of industries, including speech recognition, computer vision, and natural language processing. But in order to accomplish fast inference and training, the computational demands of these intricate deep learning models have increased dramatically, requiring the creation of specialised hardware accelerators. Although they may handle deep learning workloads, conventional general-purpose processors like central processing units (CPUs) and graphics processing units (GPUs) are not designed for the unique computational patterns and data flows that these activities require.

Numerous IT firms and academic institutions have investigated the creation of customised hardware architectures for artificial intelligence workloads as a means of addressing this difficulty. One such endeavour is Google's Tensor Processing Unit (TPU), which has a systolic array design that is tuned for the dense linear algebra operations common to deep learning models. Nevertheless, Groq, has developed the Tensor Streaming Multiprocessor (TSP) using an entirely different strategy.

The TSP is a cutting-edge processor design that rethinks a processor's basic structure and execution mechanism in an effort to solve the shortcomings of conventional processors. The TSP divides its functional units into vertical slices, each of which is devoted to a particular task, such as instruction control, memory management, integer arithmetic, floating-point arithmetic, and network interface. This is in contrast to conventional designs, which mimic homogeneous cores. When paired with a certain instruction execution model, this functional slicing technique maximises resource efficiency and reduces the overhead related to control logic and instruction decoding.[1]

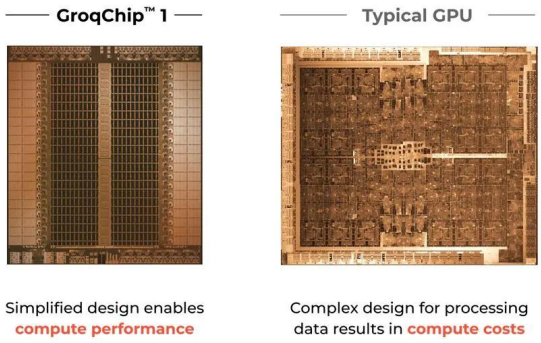


Fig. 1. **A comparison between GroqChip 1 and conventional GPU's**

The TSP chip comprises a grid of tensor streaming processors (TSPs), each capable of performing dense tensor operations on 8-bit and 16-bit data types. The key specifications of the Groq TSP include:

- TSPs arranged in a 32x32 grid[2]
- Peak performance of 1 PetaOps/sec for 8-bit operations[9]

The TSP's unique architecture allows for efficient data movement and reuse, minimizing memory bottlenecks that often limit performance in traditional architectures. This is achieved through the use of a hierarchical memory system and a decoupled access/execution model. This is known as a deterministic execution model[13].Compared to GPUs, the TSP demonstrates significant performance-per-watt advantages for AI workloads, with reported figures of up to 25x higher efficiency for certain models. This efficiency stems from the

TSP's specialized design and the elimination of unnecessary hardware components found in general-purpose processors[2].

An extensive study of the TSP architecture, its main advances, and its possible effects on the AI community are presented in this paper. A comparative analysis is also provided between Groq's TSP and Google's TPU, another well-known specialised AI accelerator, in order to illustrate the variations in architecture, performance traits, scalability, programmability, and ecosystem support between the two systems.

II. Groq TSP Architecture

Unlike traditional architectures that replicate small, homogeneous cores, the TSP employs a single, massive processor with hundreds of specialized functional units organized into vertical slices. In the context of the Tensor Streaming Processor (TSP) described in Figure 2 , The following components are present;

- INT: Integer Arithmetic Logic Unit (ALU)
  Handles integer arithmetic and logic operations.
- FP: Floating-Point Unit (FPU)
  Performs operations on floating-point numbers, which are crucial for many scientific and engineering applications.
- Data cache(D$)
  A cache memory specifically used for storing data that the processor needs to access quickly.
- Instruction Cache(I$)
  A cache memory used for storing instructions that the processor is executing, to speed up instruction fetch operations.
- Load/Store Unit(LSU)
  Manages memory operations, including loading data from memory to registers and storing data from registers to memory.
- Network Interface(NET)
  Handles the communication between different cores or processing elements, facilitating data transfer across the on-chip network.
- IF (Instruction Fetch)
  Instructions are fetched from memory and loaded into the instruction queues (ICUs) of the functional slices.
- ID (Instruction Decode)
  The instructions are decoded to determine the operations to be performed by each functional slice.
- EX (Execute)
  The functional slices (e.g., VXM (Vector Execution Module), MXM (Matrix Execution Module), SXM (Stream Execution Module)) perform the specified operations on the data streams.
- MEM (Memory Access)
  The MEM slices handle memory operations, reading data into streams or writing results back to memory.
- WB (Write Back)
  The results of computations are written back to the appropriate stream registers or memory locations.

This adaptation allows the TSP to efficiently handle deep learning workloads by leveraging a highly parallel and stream-oriented processing model.

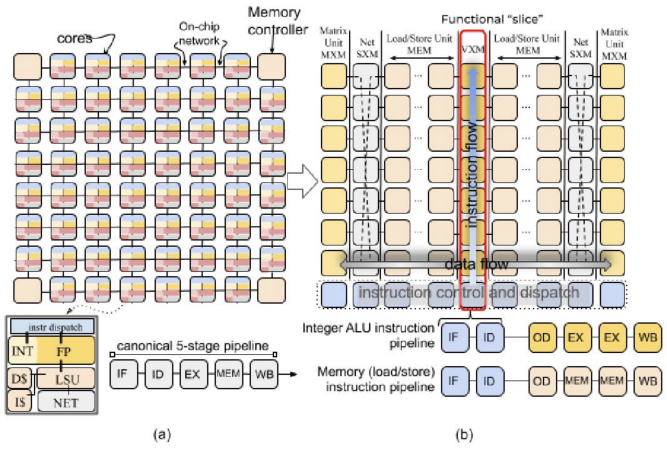


Fig. 2. **Conventional 2D mesh of cores (a) reorganized into a functionally sliced arrangement of tiles (b)**.

A. Functional Slicing

The idea of functional slicing is one of the main advances of the TSP. Rather of duplicating uniform cores, TSP reorganizes the homogeneous two-dimensional mesh of cores in Figure 2(a) into the functionally sliced microarchitecture shown in Figure 2(b). In this approach, each tile implements a specific function and is stacked vertically into a "slice" in the Y-dimension of the 2D on-chip mesh.[8] The microarchitectural design approach used in the Tensor Streaming Processor (TSP) to efficiently handle deep learning computations. This helps in high throughput by continuously streaming data through functional slices,Energy Efficiency, and a Deterministic execution The precise scheduling of instructions ensures deterministic execution, which is critical for the reliability and predictability of deep learning models.[8]

B. Instruction Execution Model

Unlike conventional CPUs, the TSP uses a unique model for executing instructions. An instruction set tailored to each functional slice's on-chip function controls it individually. For instance, the INT and FPU slices handle arithmetic operations, whereas the MEM slices support read operations only. An efficient execution flow is produced by doing away with the requirement for intricate instruction decoders and control logic using this specialised execution model.[8]

C. Memory and Data Movement

For deep learning workloads with massive tensors and high memory bandwidth needs, efficient data transport is essential. To tackle this issue, the TSP integrates an on-chip memory system with a high bandwidth and optimised data transportation algorithms. Its 80TB/s peak memory bandwidth and 220MB of on-die SRAM allow for effective data movement between functional units and memory.[3]

D. Scalability and Performance

Because of the TSP architecture's ability to grow across several processors, larger systems can be built to handle

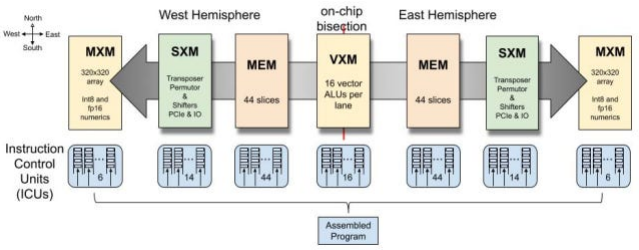


Fig. 3. **Flow of streams between the slices of TSP**

workloads that require more computing. The TSP has the ability to accelerate AI operations; according to Groq, it can achieve up to 1 PetaOp/s of INT8 performance or one quadrillion operations per second.

The TSP's primary architectural innovations, which are all focused on optimising performance and efficiency for AI inference and training workloads, are as follows: functional slicing of specialised units; a novel instruction execution model; optimised on-chip memory and data movement; and scalability across multiple chips.[3][9]

III. Deterministic Execution

One of the challenges in cloud is to reduce the tail latency to ensure not only high performance but also to meet the SLA (service level agreements). A unique opportunity from the Groq software-scheduled network and system is that execution of a given workload will be deterministic – thus, the performance variation will be negligible. Since the compile predicted execution time is a very accurate esti- mation of performance, a very tight, performance variation has been demonstrated for BERT-based inference models on a single TSP [15] as well as a distributed, four TSPs [4] Another benefit from deterministic execution is that any debugging of applications on the hardware is also simpli- fied since performance variation on real hardware is also minimized.

Hardware side-channel attacks have become problematic in CPUs and GPUs. While the shared hardware resources vary (e.g., cache, functional units, interconnect, etc.), the various attacks are fundamentally based on observing differ- ences in the execution time. As accelerators become more widely used, one can expect more effort to identify and exploit potential side channels in accelerators. However, deterministic execution or "constant" latency execution will fundamentally prevent any potential *timing* side channel attacks on the TSP architecture and enable a more secure accelerator.

IV. Potential Impact and Applications

Groq's novel TSP design has the potential to have a big influence on a lot of different areas and applications that use AI and machine learning. The TSP's primary advantages—its high performance, efficiency, scalability, and customised design for AI workloads—put it in a dominant position within the AI hardware market.

A. Real-time AI Applications

Enabling real-time AI applications, including chatbots, virtual assistants, and other interactive AI services, is one of the TSP's most exciting uses. The TSP's emphasis on quickening AI model inference, along with its low latency and high throughput, make it an excellent choice for applications that demand smooth user interfaces and quick replies.[1]

B. Large Language Model Inference

When it comes to managing large language models (LLMs) with billions of parameters, the TSP has proven to have outstanding performance. Its capacity to manage computationally demanding tasks associated with LLM inference, such text generation and natural language processing, may propel developments in fields like language translation, content creation, and conversational AI.[10]

C. Autonomous Vehicles and Robotics

The TSP is a desirable option for autonomous car applications where real-time inference and decision-making are essential because to its low power and high performance features. Furthermore, robotics applications that need for effective processing of sensor data and control algorithms may benefit from the TSP's capabilities.[1]

D. Scientific Computing and Simulations

Scientific computing applications like modelling, data analysis, and simulations may benefit from the TSP's capacity to speed up computationally demanding operations. The TSP's performance could be utilised by domains such as molecular dynamics, climate modelling, and computational fluid dynamics to expedite calculations and facilitate more intricate simulations.[1]

E. Hyperscale Datacenters and Cloud Computing

Large-scale model training and inference are typical tasks in cloud computing and hyperscale datacenter environments; these settings could benefit from the scalability and high computational throughput of the TSP. Cloud AI services that are more economical and efficient may be made possible by the TSP's design.[1]

F. Edge Computing and Internet of Things (IoT)

More potent edge computing devices may be made possible by the TSP's effective design and low power requirements, which would allow for on-device inference and lessen the requirement for cloud-based processing. This may be especially helpful for Internet of Things applications, where data privacy and low latency are critical factors.

Beyond performance records and benchmarks, Groq's TSP design has far-reaching potential effects. The TSP has the ability to stimulate innovation and open up new applications in a variety of fields, including consumer electronics, scientific research, and industrial automation. It does this by addressing the particular difficulties presented by AI workloads and providing a scalable and effective solution.[1]

V. Challenges and Future Directions

The reliability within a single TSP is designed to en- able deterministic execution; however, in order to extend such a deterministic datapath across multiple TSPs that are intercon-

nected with high speed channels, the following challenges need to be addressed.

A. Software Ecosystem Development

The creation of an extensive software ecosystem, comprising compilers, libraries, frameworks, and development tools, is one of the major obstacles to the TSP architecture's broad acceptance. Because of the TSP's unique instruction execution model and functional slicing methodology, it will take a lot of work to enable effective application development and deployment on this architecture. To build a solid software stack that is suited for the TSP, Groq will have to cultivate a thriving ecosystem and work closely with developers, researchers, and industry partners.[4]

B. Programmability and Code Portability

Functional slicing and specialised instruction execution, two of the TSP's distinctive architectural features, could make code portability and programmability difficult. When adjusting to the TSP's paradigm, developers used to conventional programming models and structures may encounter a learning curve. To handle these obstacles and guarantee code compatibility between various TSP-based systems, Groq will have to offer thorough documentation, developer tools, and training materials.[4]

C. Power Efficiency and Thermal Management

Despite the TSP's energy-efficient design, its total power efficiency and thermal properties still require careful assessment and optimisation, particularly for large-scale deployments. Ensuring optimal power efficiency and appropriate temperature management will be crucial for scalability, cost-effectiveness, and reliability as AI workloads grow more computationally demanding, especially in data centre and cloud computing environments.[3]

D. Scalability and Interconnect Performance

Large-scale TSP systems will require extensive testing and optimisation to ensure scalability and interconnect performance, even though the TSP architecture is meant to extend across numerous processors. For distributed AI workloads to execute well, numerous TSP chips and nodes must communicate and transmit data efficiently with one another. As systems scale, Groq will need to keep improving the interconnect architecture and investigating ways to reduce latency and increase bandwidth.[8][3]

E. Continued Innovation and Competition

AI hardware is a fast developing topic, with many businesses and academic organisations experimenting with new architectures and methods. To keep a competitive edge and handle new workloads and problems in the AI space, Groq will need to keep developing and improving the TSP architecture. Sustained research and development endeavours will be vital in order to meet the ever-growing computational requirements of artificial intelligence and to investigate novel application domains for the TSP architecture.

For Groq's TSP architecture to be successfully adopted and have an impact, it will be imperative to address these issues and future directions as the demand for AI grows. The TSP and other specialised AI accelerators can only be fully realised through industrial partners, researchers, software developers, hardware designers, and collaborators.[11]

VI. Comparison with Google's Tensor Processing Unit

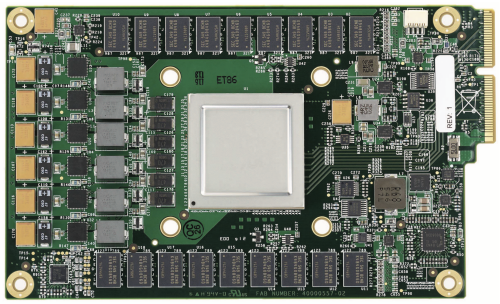


Fig. 4. **TPU printed cricuit board**

Another specialised hardware accelerator made especially for machine learning tasks is Google's Tensor Processing Unit (TPU). Although speeding up deep learning tasks is the shared goal of the TSP and TPU, their approaches to architecture and design philosophy are different. Some of their key features are listed below.

The TPU is intended to function as a coprocessor connected to a host CPU server by a PCIe link; rather of retrieving instructions on its own, it receives them from the host. A 256x256 systolic array of 8-bit integer multiply-accumulate (MAC) units serves as the primary computing unit. Partial sums are stored in a 4MB 32-bit accumulator buffer. Its units are connected by 256-byte wide datapaths and have a single 28MB on-chip memory buffer for activations and weights. Each cycle of the matrix unit yields a single 256-element vector result, with instructions flowing vertically across the functional unit slices and data flowing horizontally across each functional unit's tiles. This enables the combination of per-tile outputs within a slice, enabling efficient SIMD vector execution. [3][7]

Figure 5 depicts a TPU Block where the main computation part is the yellow Matrix Multiply unit in the upper right hand corner. Its inputs are the blue Weight FIFO and the blue Unified Buffer (UB) and its output is the blue Accumulators (Acc). The yellow Activation Unit performs the nonlinear functions on the Acc, which go to the UB.

A. Architectural Differences

The TPU takes a more conventional tack, using a matrix multiplication unit at the centre of a systolic array architecture. Deep learning workloads typically include dense linear algebra operations, which is why this design is optimised for them. [7]

On the other hand, the TSP adopts a completely distinct strategy through the use of functional slicing and a customised instruction execution architecture.[8]

B. Performance and Efficiency

For AI tasks, the TSP and TPU both make strong performance claims. However, because of the disparate architectures

Feature	Groq Tensor Streaming Multiprocessor (TSP) [3]	Google Tensor Processing Unit(TPU) [7]
Design Philosophy	Deterministic design, software-defined hardware	Domain-specific ASIC, optimized for NN inference
Execution Model	Deterministic execution, avoids out-of-order execution and caching	Deterministic execution, lacks features like out-of-order execution and multithreading
Architecture	Stream programming model, 144 independent instruction control units	Systolic array architecture, 65,536 8-bit MACs
Memory System	Large single-level scratchpad SRAM	28 MiB on-chip memory
Performance	Low-latency and high-throughput, scales predictably	92 TeraOps/second (TOPS), 15x-30x faster than contemporaries
Power Efficiency	Energy-efficient by avoiding complex features like caches	30x-80x higher TOPS/Watt than contemporary GPUs
Latency Handling	Predictable performance, optimized for response-time sensitive applications	Designed to meet 99th-percentile response-time requirements
Memory Bandwidth	No reactive components, fine-grained communication across links	Limited by memory bandwidth, but with potential for improvements with GPU memory system
Compiler Role	Orchestrates execution, controls data flow explicitly	Uses TensorFlow for workload management
Target Workloads	Wide range including CNNs, RNNs, NLP models, and dense linear algebra	Primarily NN inference, MLPs, CNNs, LSTMs
Scalability	Scales to 20K+ nodes, supports heterogeneous systems	Deployed in datacenters, optimized for cloud-scale NN workloads
Key Applications	Diverse applications like BERT, GPT-Neo, EfficientNet	Speech recognition, image recognition, language translation

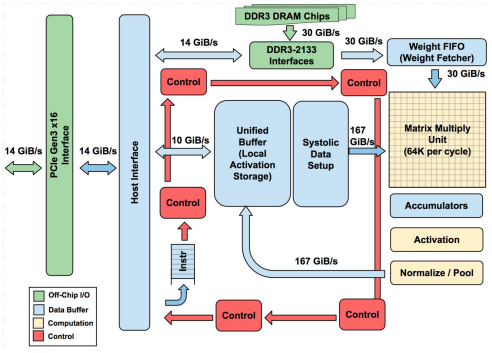


Fig. 5. **TPU Block Diagram.**

and the absence of common benchmarks, direct comparisons are difficult.

While Google's TPU v4 is said to be able to reach up to 275 TeraFLOPS of mixed-precision performance, Groq says that the TSP can deliver up to 1 PetaOp/s of INT8 performance and 250 TeraFLOPS of FP16 performance at 1.25GHz[9].By maximising resource utilisation and minimising overhead, the TSP's specialised instruction execution model and functional slicing methodology aim to improve efficiency[1]. Using a "software-defined hardware" approach, Groq's LPU architecture maximises the execution of machine learning models, improving determinism and lowering runtime uncertainty for improved speed[3].Nonetheless, the systolic array architecture of the TPU is also engineered to facilitate effective data transfer and processing for deep learning tasks.

Although there are significant performance and efficiency increases claimed by both architectures over typical GPU platforms, conclusive head-to-head benchmarking data is required to establish the relative merits of the TSP and TPU techniques.

C. Scalability and Programmability

Because of the TSP architecture's ability to spread across several chips, larger systems with even higher computational capacities can be built. Nevertheless, the TPU is made to grow by utilising several TPU chips in a single system. In terms of programmability, the TPU makes use of already-existing programming models and machine learning frameworks, like TensorFlow, which could give developers a more comfortable and familiar development environment[12]. The TSP's unique functional slicing methodology and instruction execution architecture may provide programmability and code portability issues that must be resolved by creating a strong software ecosystem.[2]

D. Ecosystem and Adoption

The fact that popular machine learning frameworks and cloud services accept Google's TPU gives it access to a wider ecosystem. Adoption and integration into current workflows can be facilitated by this ecosystem and the support of a tech behemoth like Google. Conversely, Groq, being a startup, must overcome the difficulty of creating an ecosystem—compilers, libraries, and frameworks—centered around the TSP architecture. The TSP's successful uptake and integration will rely on Groq's capacity to cultivate a thriving industry ecosystem and alliances.

VII. Conclusion

Tensor Streaming Multiprocessor by Groq is an example of a daring and avant-garde approach to creating a processor architecture suited for artificial intelligence workloads. To solve the shortcomings of conventional designs and provide better performance and efficiency for inference and training activities, the TSP tries to rethink the basic structure and execution paradigm of a processor. While speeding up deep learning workloads is a shared objective between the TSP and Google's TPU, their architectural styles, performance traits, scalability, programmability, and ecosystem support are different. A novel and promising solution is provided by the TSP's functional slicing and specialised instruction execution paradigm, but industry acceptance and success will hinge on resolving issues with software ecosystem development, power efficiency, programmability, and integration. As the demand for AI continuous to grow, both the TSP and TPU architectures

offer significant strides towards specialised hardware solutions that can unleash the full potential of deep learning and spur additional advancements in this quickly developing sector, particularly as the demand for AI grows. Both the TSP and TPU architectures offer significant strides towards specialised hardware solutions that can unleash the full potential of deep learning and spur additional advancements in this quickly developing sector, particularly as the demand for AI grows.

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